ENTRY 044 Feature Engineering Beats Volume — August 20, 2026 — Status: COMPLETE

Entry 044 — Three New Features, Same 2,288 Circuits: Floor-Collapse MAE Falls 78%

Handing the GNN Entries 032 and 034/036's Own Findings as Explicit Inputs Outperforms Every Growth Round Combined

1. What Changed

Same 2,288 circuits as Entry 043 -- only the per-node and per-edge features changed. Added three things the GNN previously had to infer purely from graph position, if it could infer them at all: (1) a control/target role flag per node, directly encoding Entry 032's proof that the control qubit's routing survival has zero effect on outcome while the target's is everything; (2) T1/T2 ratio per node, making Entry 034/036's flagged anomaly (qubit 5's unusual T1/T2 split) a single explicit number instead of something the network would have to learn to compute by dividing two raw inputs; (3) normalized route position and a final-gate flag per edge, so the model can distinguish an early swap from the entangling gate itself. Node features went from 3 to 6, edge features from 2 to 4.

2. Result: the Biggest Jump in This Project's ML Phase

	old features (Entry 043)	new features (this entry)
GNN CV MAE (all 2,288)	2.17 pts	1.05 pts
GNN CV R^2	0.869	0.976
GNN floor-collapse MAE	6.07 pts	1.34 pts
v4.1 floor-collapse MAE (unchanged)	14.67 pts	14.67 pts

Floor-collapse MAE fell from 6.07 to 1.34 points -- a 78% relative reduction, and a 91% reduction against v4.1. For comparison, growing the dataset from 1,376 to 2,198 circuits (+60% more data, Entry 042) cut floor-collapse MAE by only 15% (6.22 to 5.31). Three targeted features on the SAME data did more than five entire growth rounds combined. 1.34 points is close to the intrinsic sampling noise floor of the Aer ground-truth labels themselves (roughly 0.4-1 point at 16,384 shots per circuit) -- there may not be much further to gain on this exact task without label-level precision improvements.

Entry 044 — Conclusion

This confirms the call to prioritize features over volume. The GNN was never short on raw data relative to what THESE features needed -- it was short on being told what this project already proved. Handing a model your own findings as explicit inputs, rather than hoping message passing rediscovers them from scratch given enough examples, was the higher-leverage move by a wide margin. Growing the dataset further remains worthwhile as a background task, but this result suggests the next real gains are more likely to come from more features than from more circuits.

3. Next Step

Two directions worth checking: an ablation removing each new feature one at a time to see which of the three is actually carrying the improvement (the control/target flag is the strongest candidate, since it encodes a proven causal asymmetry rather than a correlational pattern); and a visualization of the trained network itself -- its layer structure and parameter count at the current architecture and dataset scale -- as requested.

Research Log — Index Addendum (Entry 044)

Entry	Title	Date	Status
Entry 044	Feature Engineering (control/target role, T1/T2 ratio, route position)	Aug 20, 2026	Complete

Scripts: entry044_build_graphs.py, entry044_train_gnn.py. Data: quantumbridge_data/entry044_graph_dataset.json (2,288 graphs, 6 node / 4 edge features), quantumbridge_data/entry044_gnn_folds.json, quantumbridge_data/entry044_gnn_results.json, quantumbridge_data/entry044_growth_summary.json. Floor-collapse MAE 6.07 to 1.34 points on the same 2,288 circuits -- the largest single-entry gain in the ML phase.